



(12) **United States Plant Patent**
Wright et al.

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(54) **BLUEBERRY PLANT NAMED ‘C03-038’**

(50) Latin Name: *Vaccinium corymbosum* hybrid
Varietal Denomination: **C03-038**

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patent is extended or adjusted under 35
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A01H 5/08 (2006.01)

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USPC **Plt./157**
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(58) **Field of Classification Search**

USPC **Plt./157**
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

PP12,165 P2 10/2001 Lyrene
PP20,642 P2 1/2010 Wright et al.

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(57) **ABSTRACT**

A new and distinct cultivar of blueberry (*Vaccinium corym-*
bosum hybrid) plant named ‘C03-038,’ characterized by its
semi-upright plant shape and strong plant growth vigor, low
chilling requirement, early timing of fruiting, medium fruit
size with oblate shape, firm fruit suited to handling, good fruit
flavor. This combination results in higher quality fruit with a
later availability than other varieties.

5 Drawing Sheets

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Latin name of the family, genus and species: Family—
Ericaceae. Genus—*Vaccinium*. Species—*corymbosum*
hybrid.

Variety denomination: The new blueberry plant claimed is
of the variety denominated ‘C03-038.’

BACKGROUND OF THE INVENTION

The present invention relates to a new and distinct peren-
nial variety of *Vaccinium corymbosum* hybrid (blueberry),
which has been given the variety denomination of ‘C03-038.’
The new variety ‘C03-038’ shows distinctive traits such as
early season, good flavor and firm fruit. Its market class is that
of a fruiting plant. The new variety ‘C03-038’ is intended for
use as fresh fruit for shipping, customer pick and processing
markets and as a home garden plant.

The new blueberry cultivar is a selection resulting from
seedlings produced in a controlled breeding programme of
Vaccinium varieties in Florida, USA in 2001 from a cross of
the blueberry variety known as ‘F93-368’ (seed parent) (un-
patented) and the blueberry variety known as ‘C97-390’ (pol-
len parent) (U.S. Plant Pat. No. 20,642). The seed from the
cross was sown and grown in Corindi Beach, New South
Wales, Australia. The new cultivar was discovered and
selected in 2003 as a single plant within a population of
seedlings resulted from the controlled cross, in an experimen-

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tal block in the field at Corindi Beach, New South Wales,
Australia, and has since been named ‘C03-038.’ Selection
criteria were a combination of early season, low chilling
requirement, medium fruit of good flavor, firm fruit. The new
variety was subsequently evaluated for a number of years at
the commercial farm at Corindi Beach, New South Wales,
Australia.

Asexual reproduction of the new variety ‘C03-038’ by
softwood cutting propagation since 2003 at Corindi Beach,
New South Wales, Australia has demonstrated that the new
variety reproduces true to type with all of the characteristics,
as herein described, firmly fixed and retained through succes-
sive generations of such asexual propagation, with the clones
phenotypically identical to the original plant.

The seed parent ‘F93-368’ is characterized by a medium
fruit size and very early to early timing of ripening of fruit.
The pollen parent ‘C97-390’ is characterized by a large fruit
size and very early to early timing of ripening of fruit. The
new variety ‘C03-038’ differs from the seed and pollen par-
ents in that ‘C03-038’ has a later cropping timing compared to
both parents and a smaller fruit size compared to the pollen
parent. The new variety ‘C03-038’ has maintained its distin-
guished characteristics throughout successive asexual propa-
gation.

SUMMARY OF THE INVENTION

The new blueberry variety originated from a cross of ‘F93-
368’ (seed parent) (unpatented) and the variety known as

'C97-390' (pollen parent) (U.S. Plant Pat. No. 20,642) in 2001 in Florida, USA. The seed parent is characterized by a medium fruit size and very early to early timing of ripening of fruit. The pollen parent is characterized by a large fruit size and very early to early timing of ripening of fruit.

The new blueberry variety resulted from seedlings produced in a controlled breeding program. The cross was made in 2001 in Florida, USA and the seed was sown and grown on in Corindi Beach, NSW, Australia.

The new variety was selected in 2003 from among plants located on land at Corindi Beach and has since been named 'C03-038'. Since then plants of 'C03-038' were propagated by cuttings for further evaluation and resulted to be uniform and stable. Asexual reproduction of the new variety 'C03-038' by cutting propagation since 2003 at Corindi Beach, NSW, Australia has demonstrated that the new variety reproduces true to type plants.

The new variety was selected in 2003 as a single plant within a population of seedlings resulting from controlled cross of *Vaccinium* varieties. The seedling population was planted in an experimental block in the field at Corindi Beach, NSW, Australia and the selection of the new variety took place in the same block. Selection criteria were a combination of early season, low chilling requirement, medium fruit of good flavour, elliptic fruit shape and firm fruit. The new variety was subsequently evaluated for a number of years at the commercial farm at Corindi Beach, NSW, Australia.

The following characteristics of the new variety have been repeatedly observed and can be used to distinguish 'C03-038' as a new and distinct variety of *Vaccinium corymbosum* hybrid:

1. low chilling requirement
2. semi-upright habit
3. early fruiting season
4. medium fruit size
5. good flavor
6. firm fruit
7. elliptic fruit shape

The new variety differed from the female and male parents ('F93-368' and 'C97-390' respectively) in that 'C03-038' has a smaller fruit size but firmer fruit compared to the male parent 'C97-390'. The new blueberry variety C03-038 has maintained its distinguished characteristics throughout successive asexual propagation. The variety has been repeatedly asexually reproduced through softwood cuttings in NSW, Australia and the clones are phenotypically identical to the original plant.

BRIEF DESCRIPTION OF THE DRAWINGS

The accompanying photographic illustration shows typical specimens in full color of the foliage and fruit of the new variety 'C03-038.' The colors are as nearly true as is reasonably possible in a color representation of this type.

FIG. 1 is a photograph of the new variety 'C03-038' demonstrating the plant's semi-upright growth habit.

FIG. 2 is a photograph of the fruit of the new variety 'C03-038.'

FIG. 3 is a photograph of the green fruit of the new variety 'C03-038.'

FIG. 4 is a photograph of the flowers of the new variety 'C03-038.'

FIG. 5 is a photograph of the new variety 'C03-038' growing in commercial field.

The colors in the photographs are as close as possible with the photographic and printing technology utilized. The color values cited in the detailed botanical description accurately describe the colors of the new blueberry.

DETAILED BOTANICAL DESCRIPTION

The following detailed description sets forth the distinctive characteristics of 'C03-038.' The data which defines these characteristics was collected from asexual reproductions of the original selection. Dimensions, sizes, colors, and other characteristics are approximations and averages set forth as accurately as possible. The plant history was taken on plants approximately 5 years of age, and the descriptions relate to plants grown in the field in Corindi Beach, New South Wales, Australia. Descriptions of fruit characteristics were made on fruit grown in Corindi Beach, New South Wales, Australia. Color designations are from R.H.S. Chart—edition 2007.

Classification:

Family.—Ericaceae.

Genus.—*Vaccinium*.

Species.—*corymbosum* hybrid.

Common name.—Blueberry.

Parentage:

See Parent.—F93-368 (unpatented).

Pollen parent.—'C97-390' (U.S. Plant Pat. No. 20,642).

Market class: Suitable for the Commercial, Hand Harvest Fresh, Processed, and Home Garden Markets.

PLANT

General:

Parentage.—'F93-368'×'C97-390'.

Plant height.—1.45 m.

Plant width.—1.01 m.

Growth habit.—Semi-upright.

Growth.—Low vigour.

Productivity.—Medium to high.

Cold hardiness.—Low.

Cold tolerance.—Low to medium.

Chilling requirement.—Low chill.

Tolerance to disease.—The variety is susceptible to leaf rust.

Leafing.—Medium to high. Retains leaves overwinter under the environment at Corindi Beach, NSW, Australia.

Twigginess.—High.

STEM

General:

Suckering tendency.—Low.

Mature cane color.—Between greyed orange group 165A and greyed green group 197A.

Mature cane length.—0.8 m.

Mature cane width.—9.6 mm.

Bark texture.—Smooth.

Surface texture of new wood.—Very smooth.

Internode length on strong, new shoots.—17.2 mm.

Fruiting wood.—16.2 cm in length.

FOLIAGE

General:

Time of beginning of leaf bud burst.—End of August average date 28 August, similar to ‘C97-390’.
Leaf color (top side).—Similar to green group 137B and the vein colour similar to yellow-green group N144A.
Leaf color (under side).—Yellow green group 147B.
Leaf arrangement.—Alternate.
Leaf shape.—Elliptic.
Leaf margins.—Entire.
Undulation of margin.—Absent or very weak.
Leaf venation.—Reticulate.
Leaf apices.—Acute.
Leaf bases.—Rounded.
Leaf length.—Mean 47.00 mm (45-60 mm range).
Leaf width.—Mean 28.00 mm (24-36 mm range).
Leaf length/width ratio.—1.7.
Leaf nectarines.—Absent.
Pubescence of upper side.—Absent.
Pubescence of lower side.—Absent.
Cross sectional profile.—Flat.
Longitudinal profile.—Straight.
Attitude.—Horizontal.

Petioles:

Length.—Mean 3.1 mm (2-4 range).
Width.—1.9 mm.
Color.—Yellow green group 145A.

FLOWERS

General:

Time of beginning of flowering.—Early.
Time of 50% anthesis.—Average date 20 July.
Flower shape.—Urceolate.
Flower bud density.—Medium to dense.
Flower fragrance.—Absent.
Flower arrangement.—Alternately.
Flower type.—Complete flower, having sepals, petals, stamens and pistils.

Corolla:

Color.—Green-white group 157D.
Length.—6.6 mm.
Width.—Mean 7.1.
Aperture width.—5.1 mm.
Anthocyanin coloration of corolla.—Absent or very weak.
Corolla ridges.—Present.
Protrusion of stigma.—Present.
Corolla shape.—Urceolate.

Inflorescence:

Length.—22-31 mm.
Diameter.—15-22 mm.
Length of peduncle.—17 mm.
Surface texture of peduncle.—Smooth.
Color of peduncle.—Yellow green group 147D.
Length of pedicel.—4.7 mm.
Surface texture of pedicel.—Smooth.
Color of pedicel.—Yellow green group 147D.
Number of flowers per cluster.—4.
Flower cluster density.—Sparse.

Calyx (with sepals):

Diameter.—6.11 mm.

Stamen:

Length.—6.09 mm.
Number per flower.—10.
Filament color.—Yellow-green group 145D.
Style.—Length — 8.3 mm.
Color.—Yellow-green group 144C.

Pistil:

Length.—11.3 mm.
Ovary color (exterior).—Green group 143A.

Anther:

Length.—3.06 mm.
Number.—10.
Color.—Greyed-orange group n167B.

Pollen:

Abundance.—Abundant.
Color.—Yellow group 4D.
Self-compatibility.—Possible, but smaller fruit sizes and very few seeds.

FRUIT

General:

Time of fruit ripening.—Early season.
Time of 50% maturity.—Average date 18th September.
Fruit development period.—60 days.
Cluster density.—Sparse.
Numbers of berries per cluster.—4.
Unripe fruit color.—Yellow-green group 144A.
Ripe berry color.—Violet-blue group 98B.
Berry surface wax abundance.—Strong to very strong.
Berry flesh color.—Yellow-green group 145C.
Berry weight.—Mean 1.6 g (1.2-2.5 range).
Berry height from calyx to scar.—13.7 mm.
Berry diameter.—14.01 mm.
Berry shape.—Elliptic.
Fruit stem scar.—Very small and dry.
Sweetness when ripe.—Medium to high.
Firmness when ripe.—Very firm.
Acidity when ripe.—Medium.
Storage quality.—Long storage.
Suitability for mechanical harvesting.—Not suitable.
Self-fruitfulness.—High.
Uses.—Fresh.

SEED

General:

Seed abundance in fruit.—Medium (166 in 10 fruit).
Seed color.—Greyed-orange group 165B.
Seed length.—1.94 mm.

COMPARISON BETWEEN PARENTAL AND COMMERCIAL CULTIVARS

		Variety	Comparator variety				
60	Character- istic	C03- 038	C99-042	Emerald	C97- 390	Star	Snow- chaser
	Soluble solid content (%)	11.2	13.1	11.6	12.2	13.1	14.3
	Titratable acidity (%)	0.6	0.3	0.5	0.3	0.4	0.5

-continued

Character- istic	Variety	Comparator variety				
	C03-038	C99-042	Emerald	C97-390	Star	Snow-chaser
Fruit weight (g)	1.6	1.9	3.0	2.0	1.9	1.7
Fruit shape	elliptic	oblate	oblate	oblate	oblate	oblate
Firmness (g/mm)	300	273	187	211	210	188
Leaf length (mm)	47	52.5	62	—	65.7	56.1
Leaf width (mm)	28	24.6	37	—	27.7	32.1

-continued

Character- istic	Variety	Comparator variety				
	C03-038	C99-042	Emerald	C97-390	Star	Snow-chaser
Plant habit	Semi-upright	Semi-upright to spreading	Spread-ing	Semi-upright	Strongly upright to upright	Semi-upright
Time of fruit ripening	Early	Early	Mid to late	Early	Late	Very early

It is claimed:
1. A new and distinct variety of blueberry plant named ‘C03-038,’ substantially as illustrated and described herein.

* * * * *

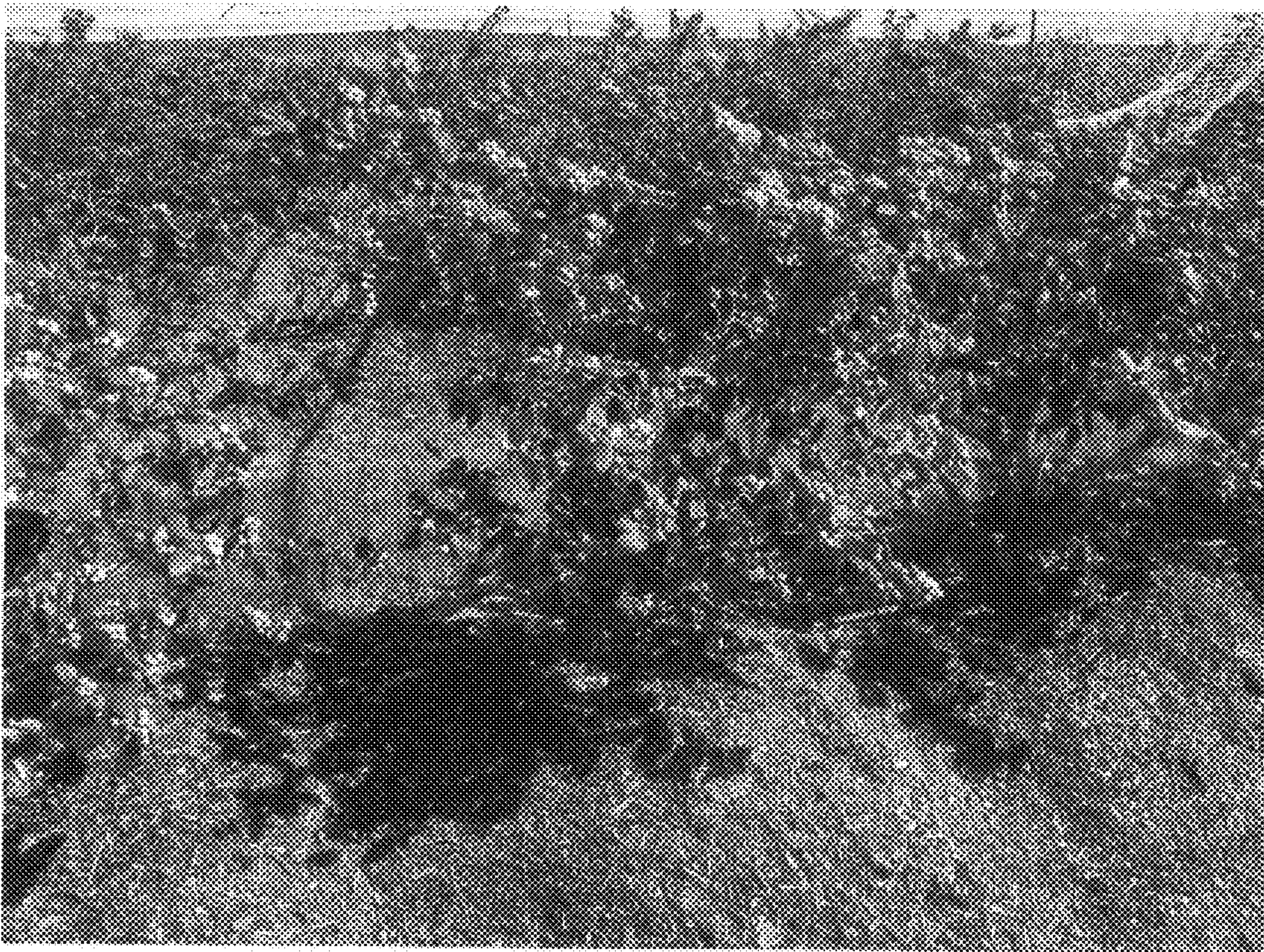


FIG.1



FIG.2



FIG.3

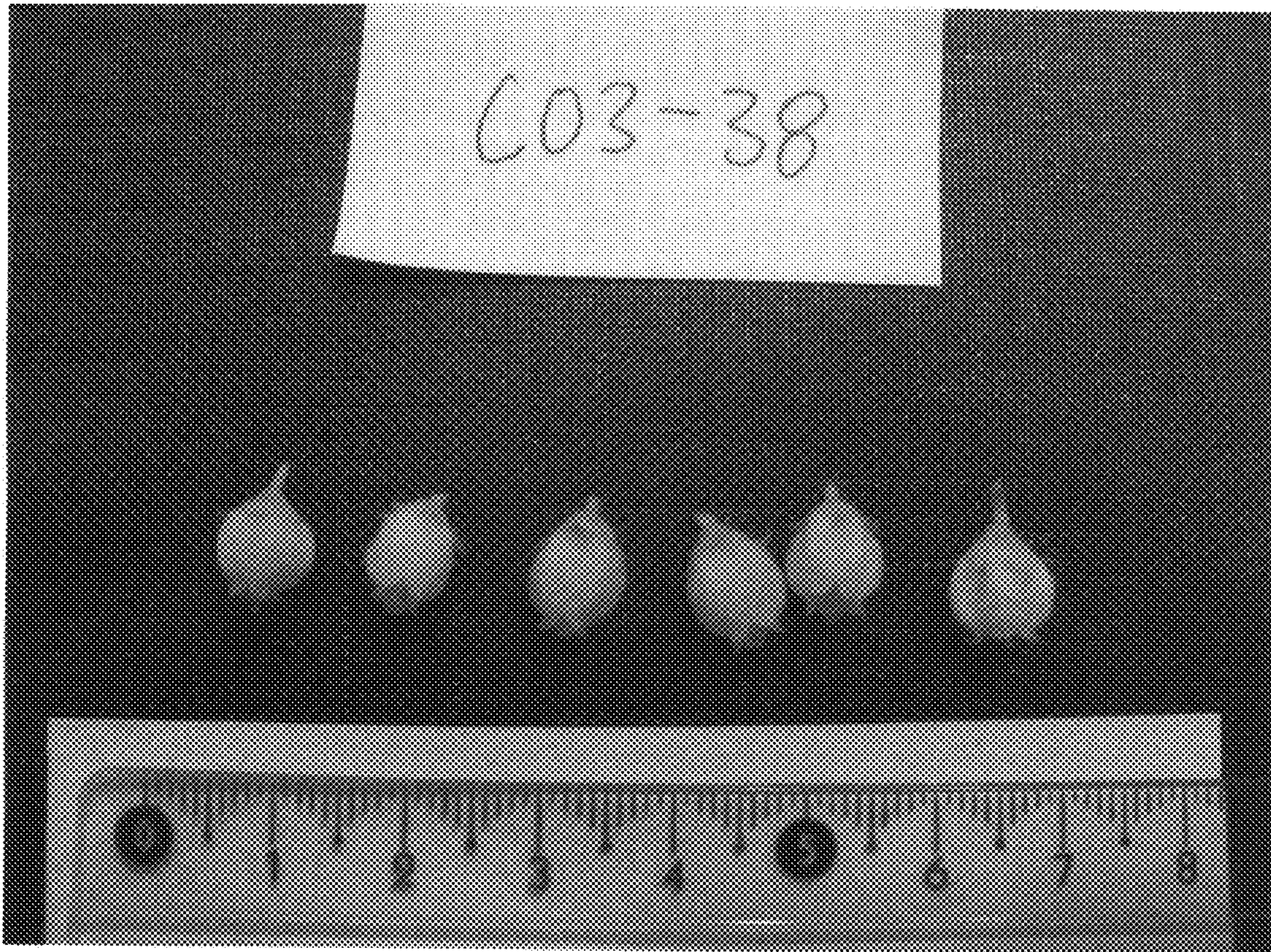


FIG.4



FIG.5